

Video Link

<https://youtu.be/FqgJusmUy5I>

Peak Performance Partners

Hayden Miller, Saahas Pulivarthi,
Nicole Roy, Olivia Shoaf,
Mason Bozel



ASA DATAFEST

*"When things are looking bleak,
always choose Peak"*

Issue at Hand



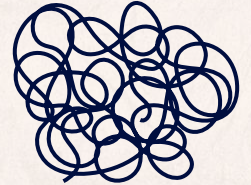
Task

Stormont Vail Health (SVH), a nonprofit integrated health system based in Kansas, has asked us to analyze a set of real patient data and find ways to improve the patient journey through their system.



Patient Journey

A patient's journey begins when the patient seeks medical care, along with all visits centered on that diagnosis. Improvements could include limiting appointment gaps, amount of testing, and the amount of money spent on giving care.





Approach



Data:

- Visit type, emergency status, social determinants, length of patient journey.
- We cleaned the data by removing rows with missing values, and combined the patient related datasets into one.

Goal:

- Focus on one area of the data and provide evidence based recommendations for improvement.

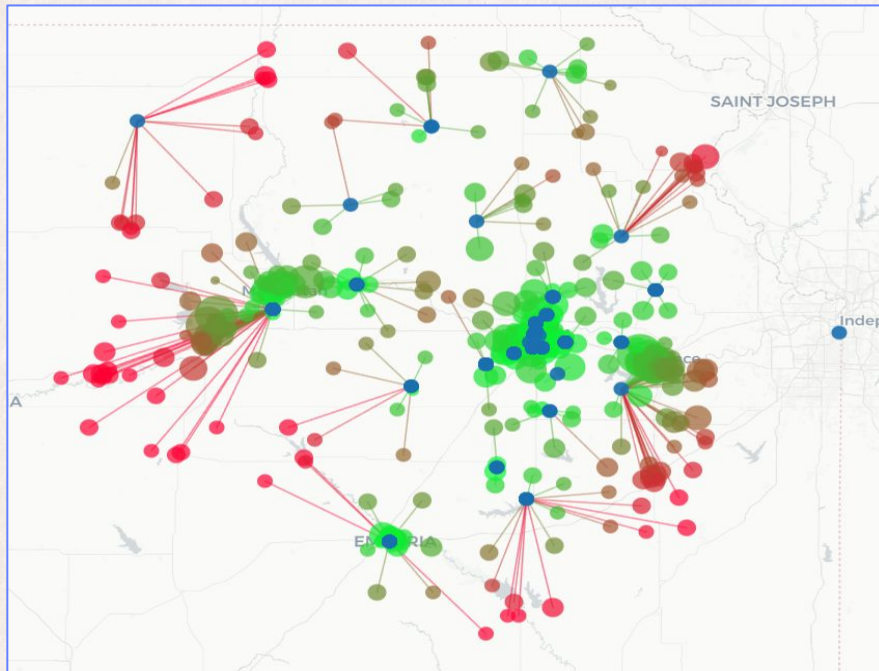
Analysis and Tools:

- Stratified Group Summaries, GLM & Linear Regression, Significance Testing, and Geospatial Mapping using R, Python, and Excel.

Research:

- We decided to **focus on transportation** as our area of attention, as about 30% of Kansas' population lives in rural areas according to *America's Health Rankings* ([Source](#)), and people who live in rural areas are more likely to experience transportation burdens, according to the *Journal of Transport Geography* ([Source](#)).
-

First Glance



64.5%

of the patient dataset did not
have a clear indicator of precise
location

8.8 miles

is the mean distance to the
nearest hospital

More than 30%

live more than 10 miles away from
the nearest hospital

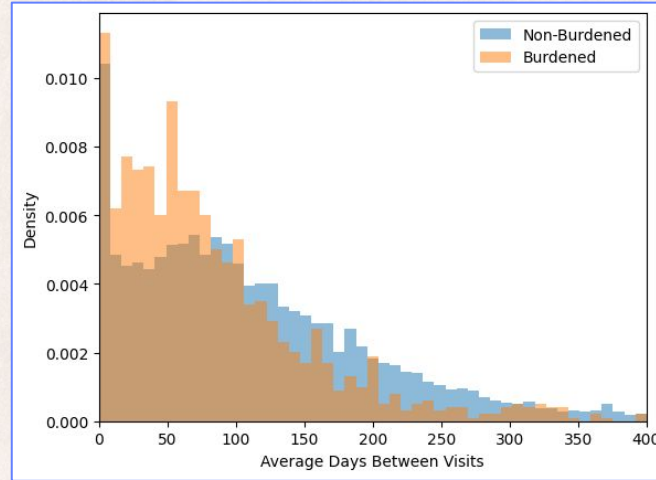
Burdened Patient Outlooks

Burdened patients have more frequent hospital visits

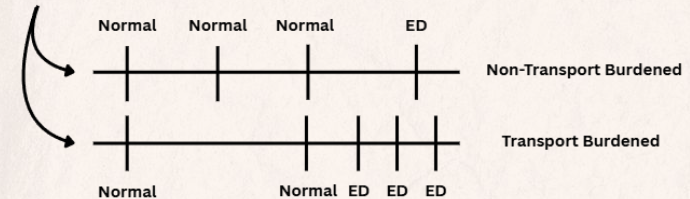
Burdened patients on average visit the hospital about every 88 days, while non-burdened patients visit every 117 days.

However, burdened patients got to the ED 3.2x more often

This indicates a potential issue with preventative care for burdened patients. ED's are also more expensive and require more resources.

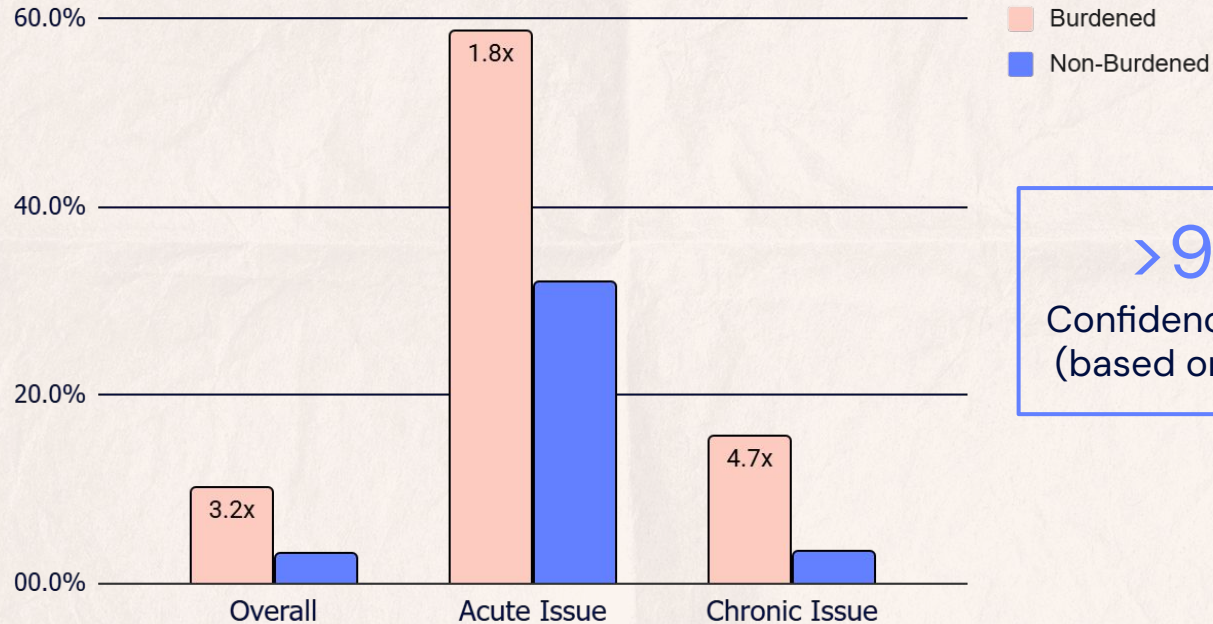


Issue Start



Insights

ED Rates for Burdened & Non-Burdened Patients



>99.99%

Confidence for rates listed
(based on stratified data)

Further Investigation



Cumulative Burden → ED Reliance

- **Logistic regression** predicting ED visits from **transportation burden**, controlling for age, sex, race, smoking, financial strain, partner violence, and housing stability.
- **Transportation burden** is the **single strongest predictor of ED visits**:
OR = 2.88x ($p = 0.00097$).
- **Additional social barriers alone increase ED risk**, but among transport burdened patients they actually reduce ED utilization.

Frequent Visits, Wrong Setting

- **OLS regression**, controlling for age, race, smoking.
- Transport-burdened patients:
44 fewer days between visits
($p = 0.019$).
- Combined with 3.2x ED rate → burdened patients aren't missing care, they are getting care in the most expensive way.



Recommended Solutions:

Enhanced Outreach to Susceptible Patients

- Patients who self-identify as transportation burdened as well as any one or more of the social determinant factors that were significant should be flagged for enhanced outreach.
 - These patients should be recommended for more frequent preventative care visits to identify problems before they become life-threatening.
- In the long run, preventative care is cheaper, minimizes hospitalizations, and requires less resources.



Recommended Solutions:

Telehealth Implementation

- Allows for less immediately life-threatening issues to be addressed.
- Symptoms of chronic illnesses can be described often without physical observation, and further visits can occur only when necessary.
- Telehealth appointments are much cheaper and use less resources.
 - Requires less staff, shorter, less no shows, faster patient turnover.
 - Most importantly, does not require any travel.
- Internet is expanding, making telehealth more widely available.
 - The percentage of rural areas with minimal to no broadband access has been cut by two thirds in the years leading up to 2025, according to the *Washington Post* ([Source](#)).